Building Advanced Agentic Systems on AWS

MODULE 01: Multi-Agent Architecture and Communication Patterns

- Strands Agents as Tools pattern
- Deploy MCP servers in AgentCore Runtime
- Deploy A2A servers in AgentCore Runtime
- AgentCore Runtime service contract
- Strands Multi-agent Orchestration Patterns

MODULE 02: Context Engineering and Performance Optimization

- Strands Agents - Caching (Provider Agnostic)
- Prompt Caching – supported models and limits
- Prescriptive Guidance – Context Engineering
- Anthropic – Effective Context Engineering for AI Agents

MODULE 03: Security and Compliance Implementation

- Agentic AI Security Scoping Matrix
- How session binding in AgentCore Identity works
- Endpoint policy for both IAM and OAuth users
- OWASP Top 10 for Agentic Applications for 2026

MODULE 04: Production Monitoring, Observability, and Evaluation

- On-demand Evaluation Code Sample
- How Amazon uses AI agents to support compliance screening of billions of transactions per day

MODULE 05: Well-Architected Agentic AI Systems

- AWS Well Architected Framework